

Smart City Complaint Analysis Report

Problem Statement

Modern cities receive thousands of citizen complaints related to sanitation, roads, water supply, traffic, drainage and public safety. This project analyzes complaint data to identify recurring issues, service inefficiencies and operational priorities.

Objective

Build a Smart City Complaint Analytics Dashboard to analyze complaint trends, service performance, complaint hotspots and citizen satisfaction.

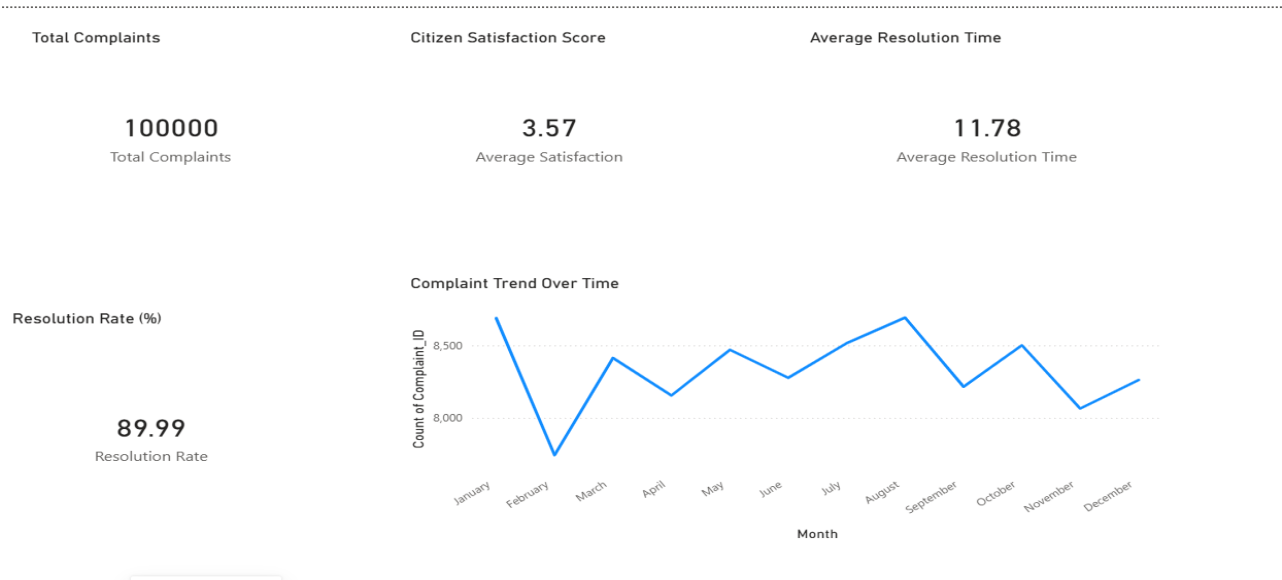
Dataset Description

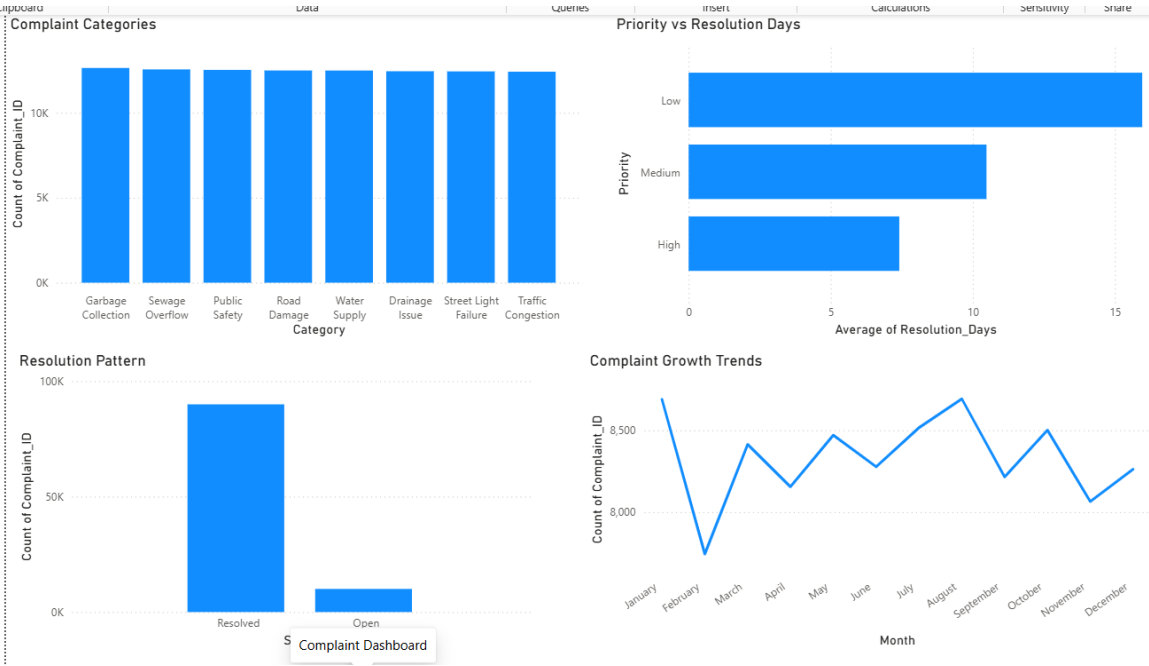
Dataset includes Complaint ID, Complaint Date, Resolved Date, Zone, Category, Department, Priority, Status, Satisfaction Score, Description, Resolution Days and Severity Score.

KPI Definitions

Total Complaints, Resolved Complaints, Resolution Rate, Average Resolution Time, High Priority Complaints, Citizen Satisfaction Score, Complaint Hotspot Zones and Department Performance Score.

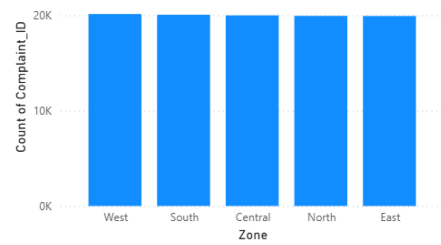
Dashboard Visualizations



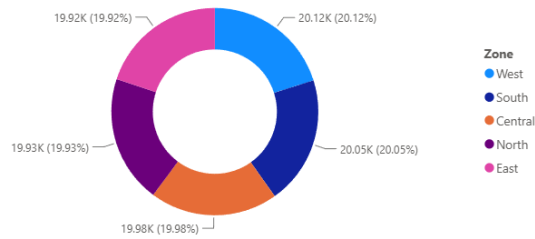


Complaint Dashboard

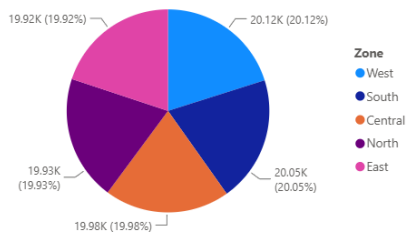
Zone Wise Issues



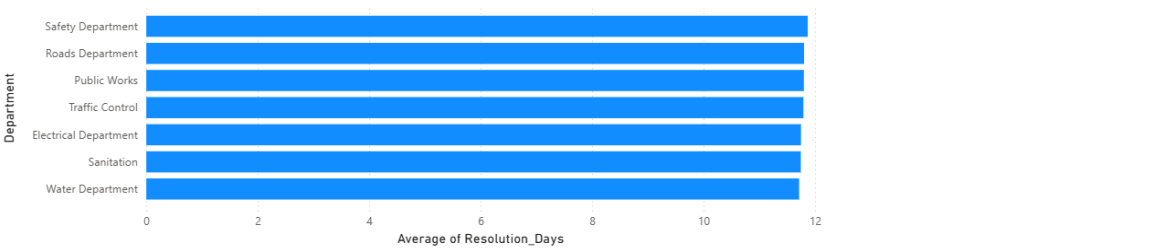
Complaint Hotspots



Geographic Distribution of Complaints

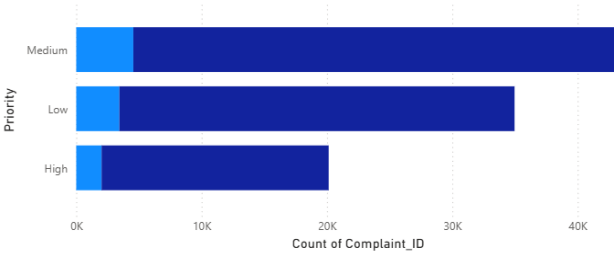


Department Performance

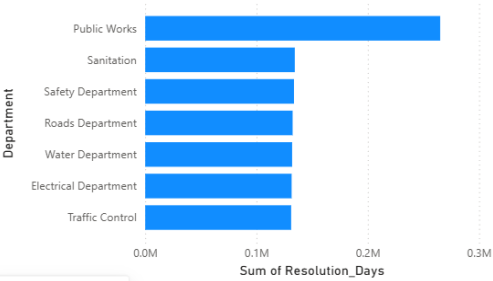


Priority handling

Status ● Open ● Resolved

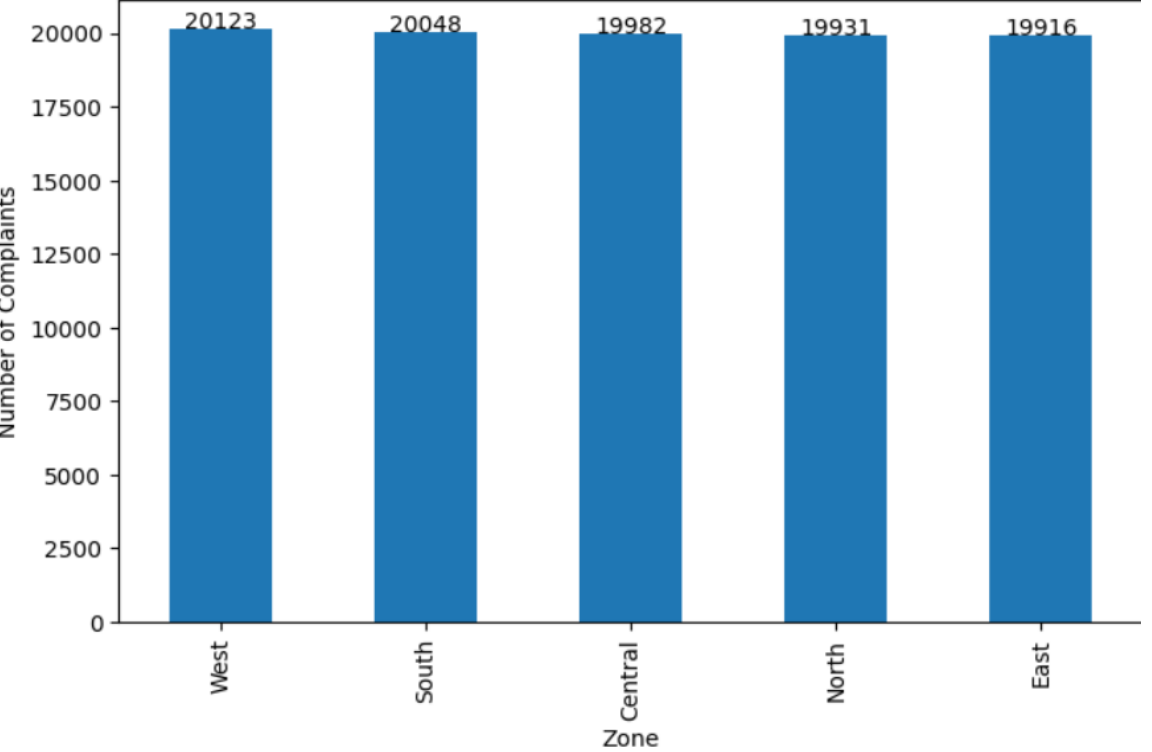


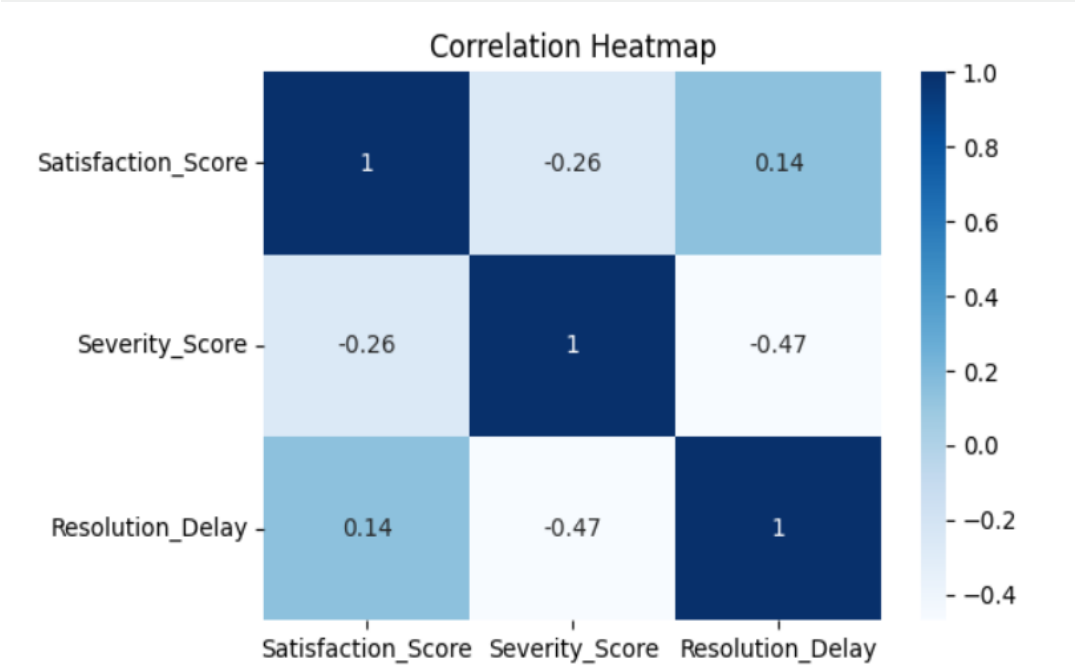
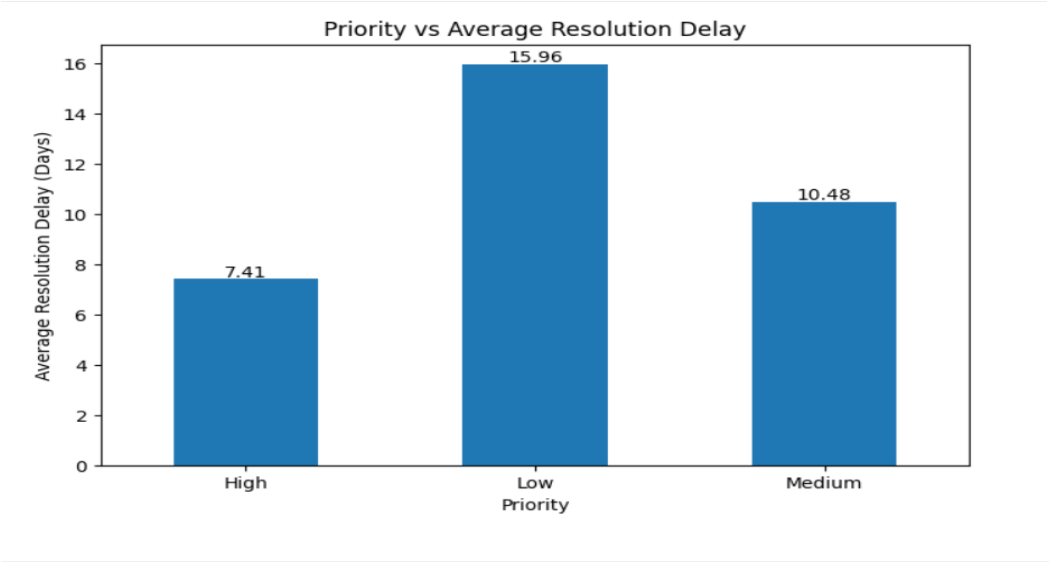
Resolution Efficiency by Department

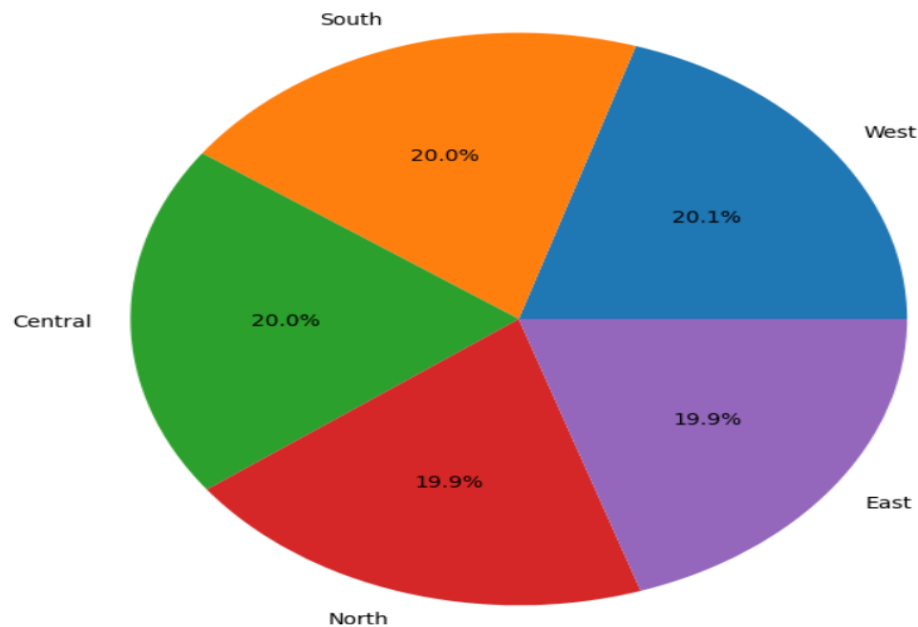


Service Performance

Zone vs Complaint Count







Business Insights

1. Frequently occurring complaint categories highlight key civic challenges.
2. Complaint hotspot zones indicate recurring urban problems.
3. Departments with higher average resolution time require process improvement.
4. Long resolution times and unresolved complaints reduce citizen satisfaction.
5. Authorities should prioritize high-priority complaints, hotspot zones and underperforming departments.

1. Which complaint categories occur most frequently?

The most frequently occurring complaint categories were related to essential civic services such as garbage collection, road damage, sewage overflow, and water supply issues. These categories accounted for a significant share of total complaints, indicating areas where city infrastructure and service delivery require continuous monitoring and improvement.

2. Which areas experience recurring civic problems?

Complaint hotspot analysis revealed that certain zones consistently generated higher complaint volumes than others. These recurring complaints suggest persistent infrastructure or service issues in those areas. Authorities should prioritize these zones for preventive maintenance and resource allocation.

3. Which departments take the longest to resolve complaints?

Department performance analysis showed that some departments had higher average resolution times compared to others. Longer resolution times may indicate operational bottlenecks, staffing limitations, or increased workload. Improving workflow efficiency and resource management can help reduce delays.

4. What factors reduce citizen satisfaction?

Citizen satisfaction was negatively impacted by:

- Long complaint resolution times
- High numbers of unresolved complaints
- Delays in handling high-priority issues
- Repeated complaints from the same zones
- Inefficient service delivery by overloaded departments

5. What actions should city authorities prioritize?

Based on the analysis, city authorities should prioritize:

- Faster resolution of high-priority complaints
- Immediate attention to complaint hotspot zones
- Improvement of departments with low performance
- Preventive maintenance in areas with recurring issues
- Continuous monitoring of citizen satisfaction scores
- Data-driven resource allocation to overloaded departments

Key Findings

Complaint distribution varies across zones. Some departments resolve complaints faster than others. High-priority complaints require special monitoring. Complaint trends help identify recurring issues.

Recommendations

Improve response times, allocate resources to overloaded departments, monitor hotspot zones, strengthen preventive maintenance and implement priority-based complaint management.

Innovation

Complaint Severity Scoring, Recurring Issue Detection, Department Performance Benchmarking and Citizen Satisfaction Analytics.

Future Scope

AI-based complaint categorization, predictive analytics, GIS mapping, real-time monitoring and automated alerts for high-priority complaints.

Conclusion

The Smart City Complaint Analytics Dashboard enables authorities to identify recurring civic problems, monitor service performance, and make informed decisions. By focusing on high-priority complaints, hotspot zones, and departmental efficiency, city leadership can improve urban service delivery and citizen satisfaction.